AWS STUDENT COMMUNITY DAY · ISLAMABAD

FROM CODE TO CLOUD

Your first DevOps pipeline on AWS

FarhanAshrafDev · Beginner session + live demo

CLOUD IN 30 SECONDS

Rent building blocks. Control them by API.

0 servers to buy before your first deployment

On demand

Create what you need, when you need it.

Programmable

Use code instead of repeating console clicks.

Quick show of hands

01

Who has written any code?

02

Who has deployed it to the internet?

03

Who knows what DevOps actually means?

No previous DevOps knowledge is required today.

THE PROBLEM BEFORE THE TOOLS

“It works on my machine.”

> **How does every change reach users quickly, safely, and repeatedly?**

Manual deployment depends on memory. Reliable delivery depends on a system.

DEV + OPS + FEEDBACK

DevOps is a way of working

Culture

Builders and operators own the result together.

Practices

Small changes, review, measurement, learning.

Automation

Machines repeat the boring and risky steps.

Before vs. with DevOps

Manual

Large releases
Different steps
Slow feedback
Hero debugging

Repeatable

Small releases
Automated checks
Fast feedback
Shared ownership

Speed comes from confidence.

THE DEVOPS LOOP

One change. Five questions.

1 · CODE — what changed?

NEXT

2 · BUILD — what will we ship?

NEXT

3 · CHECK — is it safe enough?

NEXT

4 · DEPLOY — how does it go live?

NEXT

5 · OBSERVE — did it work?

CI/CD WITHOUT THE ALPHABET SOUP

Integrate early. Deliver repeatedly.

Continuous Integration

Combine small changes and check each one.

Continuous Delivery

Keep a checked version ready to release.

Continuous Deployment

Automatically release a change that passes.

The goal is fast feedback—not a fancy pipeline diagram.

INFRASTRUCTURE AS CODE

The environment is also code

Application code

What users see and use.

Pipeline code

How every change is checked and delivered.

Infrastructure code

Which AWS resources, permissions, and settings exist.

Today, one CloudFormation template describes the whole AWS environment.

CLICKOPS VS. INFRASTRUCTURE AS CODE

One template. Same result.

Console memory

Click 30 times
Forget one setting
Hard to review
Hard to repeat

CloudFormation

Version in Git
Review the diff
Deploy consistently
Delete as a stack

The template is our blueprint.

SKILLS REQUIRED

Start small.
Learn in layers.

Start today

Basic Git
Read simple code
Curiosity
Willingness to break things

Build next

Linux basics
AWS + IAM
CI/CD
IaC + monitoring

Not required: Docker, Kubernetes, EKS, Argo CD, or AI.

ONLY SIX AWS BUILDING BLOCKS TODAY

Each tool has one simple job

CloudFormation
Create the environment

CodeConnections
Connect GitHub once

CodePipeline
Coordinate the stages

CodeBuild
Check and package

Amazon S3
Store the webpage

CloudFront
Deliver it securely

TODAY'S ARCHITECTURE

Git push to live website

GITHUB PUSH

↓ CODECONNECTIONS

CODEPIPELINE · SOURCE / CHECK / DEPLOY

↓ CHECKED WEBSITE FILES

PRIVATE AMAZON S3

↓ SECURE DELIVERY

CLOUDFRONT → WEB BROWSER

ONE STACK + ONE HUMAN HANDSHAKE

Setup once. Push forever.

1 · Deploy the CloudFormation stack

NEXT

2 · Authorize the pending GitHub connection

NEXT

3 · Push changes and watch the pipeline

Human decides

Which repository AWS may access.

Automation repeats

Every later source, check, and deploy step.

LIVE-DEMO MISSION

Change three values. Ship them live.

- OK Show version 1 in the browser
- OK Edit message, version, and color
- OK Commit and push to GitHub
- OK Watch Source → Check → Deploy
- OK Refresh and reveal version 2

Audience prediction: will all three stages go green?

NO SLIDES FOR THE NEXT 8 MINUTES

LIVE
DEMO

Change / Commit / Pipeline / Website

WHAT JUST HAPPENED?

The pipeline did the checklist

- **CodeConnections detected the GitHub push**

NEXT

- **CodePipeline downloaded the commit**

NEXT

- **CodeBuild checked and packaged demo/**

NEXT

- **The deploy stage updated private S3**

NEXT

- **CloudFront served the new page**

FAILURE AND SECURITY ARE PART OF DEVOPS

Stop early. Limit access.

Check fails?

Deployment stops before users see the change.

Need recovery?

Revert the Git commit and run the same path.

GitHub access?

One approved app connection—no access key in Git.

AWS access?

Service roles receive only the permissions they need.

A red pipeline is useful feedback delivered early.

YOUR FIRST 30 DAYS

Learn by shipping

WEEK 1 · Git + Linux basics

NEXT

WEEK 2 · AWS IAM, S3, and networking

NEXT

WEEK 3 · CodeBuild + CodePipeline

NEXT

WEEK 4 · CloudFormation + CloudWatch

Build one tiny project. Automate it. Break it. Recover it. Explain it.

REMEMBER ONLY THREE THINGS

Small. Automated. Observable.

01 · Small changes
Reduce risk and simplify recovery.

02 · Automated path
Make delivery repeatable.

03 · Fast feedback
Learn before users suffer.

SLIDES · WEBPAGE · ONE-STACK TEMPLATE · RUNBOOK

Thank you.
Questions?

Fsociety-Bulc / AWSSCDWorkshop
merge a PR, watch it deploy

Deploy link, cleanup, and rehearsal steps are in the repository.